Python
Conditional Statements

Objective

- Python Indentation
- Decision making
  - if statement
  - if..else statements
  - nested if statements
  - if-elif ladder
  - Short Hand if statement
  - Short Hand if-else statement
- Summary

Python Indentation

Most of the programming languages like C, C++, and Java use braces { } to define a block of code. **Python, however, uses indentation**.

A code block (body of a function, loop, etc.) starts with indentation and ends with the first unindented line. The amount of indentation is up to you, but it must be consistent throughout that block.

Generally, four whitespaces are used for indentation and are preferred over tabs. Here is an example.

```
if True:
    print('Hello')
    a = 5
```

Decision making

- Decision making statements in programming languages decides the direction of flow of program execution. Decision making statements available in python are:
- if statement
- if..else statements
- nested if statements
- if-elif ladder
- Short Hand if statement
- Short Hand if-else statement

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if statement

- It is used to decide whether a certain statement or block of statements will be executed or not i.e if a certain condition is true then a block of statement is executed otherwise not.
- We can use condition with bracket () also.

Syntax:

```
if condition:
  # Statements to execute if
  # condition is true
```

```
if condition:
    statement1
statement2
```

- # Here if the condition is true, if block will consider only statement1 to be inside its block.

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Example

```
i = 10
if i > 15:
  print (“i is less than 15")
print ("I am Not in if")
```

Output:

```
I am Not in if
```

As the condition present in the if statement is false. So, the block below the if statement is not executed

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if..else statements

- The if statement alone tells us that if a condition is true it will execute a block of statements and if the condition is false it wont. But what if we want to do something else if the condition is false. Here comes the else statement.
- We can use the **else statement with if statement** to execute a block of code when the condition is false.

Syntax:

```
if (condition):
    # Executes this block if
    # condition is true
else:
    # Executes this block if
    # condition is false
```


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Example:

```
i = 20;
if (i < 15):
    print ("i is smaller than 15")
    print ("i'm in if Block")
else:
    print ("i is greater than 15")
    print ("i'm in else Block")
print ("i'm not in if and not in else Block")
```

Output:

```
i is greater than 15
i'm in else Block
i'm not in if and not in else Block
```


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nested if statements

- Nested if statements means an if statement inside another if statement or else statement.

Syntax:

```
if (condition1):
  # Executes when condition1 is true
  if (condition2):
      # Executes when condition2 is true
  # inner if Block is end here
# outer if Block is end here
```


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Example

```
i = 10
if (i == 10):
    #  First if statement
    if (i < 15):
        print ("i is smaller than 15")
    if (i < 12):
        print ("i is smaller than 12 too")
    else:
        print ("i is greater than 15")
```

Output:

```
i is smaller than 15
i is smaller than 12 too
```

if-elif ladder

- User can decide among multiple options. The if statements are executed from the top down.
- As soon as one of the conditions controlling the if is true, the statement associated with that if is executed, and the rest of the ladder is bypassed.
- If none of the conditions is true, then the final else statement will be executed.

Syntax:-

```
if (condition):
   statement
elif (condition):
   statement
else:
   statement
```


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Example

```
i = 20
if (i == 10):
    print ("i is 10")
elif (i == 15):
    print ("i is 15")
elif (i == 20):
    print ("i is 20")
else:
    print ("i is not present")
```

Output:

```
i is 20
```


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Short Hand if statement

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Whenever there is only a single statement to be executed inside the if block then shorthand if can be used. The statement can be put on the same line as the if statement.

Syntax:

```
if condition: statement
```

Example:

```
# Python program to illustrate short hand if
i = 10
if i < 15: print("i is less than 15")
```

Output:

```
i is less than 15
```


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Short Hand if-else statement

- This can be used to write the if-else statements in a single line where there is only one statement to be executed in both if and else block.

Syntax:

```
statement_when_True if condition else statement_when_False
```

Example:

```
# Python program to illustrate short hand if-else
i = 10
print(True) if i < 15 else print(False)
```

Output:

```
True
```


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Thank You